

CURSO 1

MODULO 1

Modo de revisión

This screenshot shows the 'About this module' page for 'Introduction to Data Concepts'. The left sidebar contains a 'COURSE OVERVIEW' section with a list of topics: 'About this course', 'Earn a credential!', 'Course design', and 'MODULE 1: TYPES OF DATA'. Under 'MODULE 1: TYPES OF DATA', the following topics are listed: 'About this module' (checked), 'Why is data important?' (checked), 'How much data is there?' (checked), and 'Data can be structured or unstructured' (checked). The main content area has a blue header with 'Lesson 4 of 27' and 'EXIT COURSE'. Below the header, the title 'About this module' is displayed. The main content area is dark blue with the text 'Module 1 Types of data' and the IBM SkillsBuild logo.

This screenshot shows the 'Data Fundamentals' page. The left sidebar is identical to the previous screenshot. The main content area has a blue header with 'EXIT COURSE'. Below the header, the title 'Data Fundamentals' is displayed. The main content area is white with the text 'This course is one of several in the Data Fundamentals series. Complete all of the courses in the Data Fundamentals learning plan and earn the Data Fundamentals digital credential from IBM!'. To the right, there is a large pink and purple graphic with the text 'Data Fundamentals' and the IBM SkillsBuild logo. At the bottom, there is a link to 'Data Fundamentals' on Credly.

Modo de revisión

This screenshot shows the 'Reflect' dialog box. The left sidebar is identical to the previous screenshots. The main content area is white with the title 'Reflect'. Below the title, there is a text input field with the text 'La conclusión que se puede extraer sobre los datos estructurados en la tabla dependerá de la información presentada. Algunos puntos clave a considerar pueden incluir patrones, tendencias, relaciones entre variables y posibles anomalías en los datos.' Below the text input field, there is a 'Remaining characters: 251' indicator. At the bottom, there are navigation buttons: a back arrow, a forward arrow, and a 'Reflect' button.

Modo de revisión

COURSE OVERVIEW

About this course

Earn a credential!

Course design

MODULE 1: TYPES OF DATA

About this module

Why is data important?

How much data is there?

Data can be structured or unstructured

Databases

Think of data as qualitative or quantitative

Quantitative data relates to numbers.

Quantitative data:

- Is also called **numerical data**
- Represents things that can be measured and assigned values
- Can be counted and measured, such as height, weight, length, blood pressure, the temperature outside, and so on

Qualitative data relates to words and descriptions.

Qualitative data:

- Is also called **categorical data**
- Represents the characteristics, attributes, properties, and qualities of things
- Describes data using language (rather than numbers), such as smell, location, color, texture, marital status, and so on

Take a look at a quick scenario to help you understand the difference between quantitative and qualitative data.

EXIT COURSE

Introduction to Data Concepts

100% COMPLETE

quantitative

Quiz

MODULE 2: BIG DATA

About this module

What is big data?

The 5 V's

Quiz

MODULE 3: TYPES OF DATA ANALYTICS

About this module

Data analytics answers questions

Quiz Results

Your score 100%

PASSING 75%

TAKE AGAIN

EXIT COURSE

MODULO 2

Introduction to Data Concepts

100% COMPLETE

quantitative

Quiz

MODULE 2: BIG DATA

About this module

What is big data?

The 5 V's

Quiz

MODULE 3: TYPES OF DATA ANALYTICS

About this module

Data analytics answers questions

Lesson 11 of 27

What is big data?

Many people interact daily with an increasing number of internet-connected devices that record data, from text messages and school records to travel habits and movie preferences. In a digital world, everyone that uses devices like these leaves a trail of data. The term for this kind of data is **big data**.

The term "big data" has been around for years, but there isn't a universal, official definition of big data. Here are some definitions to get you thinking.

Expand each section to review each definition.

EXIT COURSE

Introduction to Data Concepts

100% COMPLETE

quantitative

Quiz

✓

MODULE 2: BIG DATA

About this module

✓

What is big data?

✓

The 5 V's

✓

Quiz

✓

MODULE 3: TYPES OF DATA ANALYTICS

About this module

✓

Data analytics answers questions

✓

Quiz

✓

Lesson 12 of 27

EXIT COURSE

The 5 V's

The elements of big data can be explained using five broad characteristics called the **5 V's**. The 5 V's are **Volume**, **Variety**, **Velocity**, **Veracity**, and **Value**. The 5 V's help data scientists make sense of what they are working with and are depicted in this diagram.

Introduction to Data Concepts

100% COMPLETE

quantitative

Quiz

✓

MODULE 2: BIG DATA

About this module

✓

What is big data?

✓

The 5 V's

✓

Quiz

✓

MODULE 3: TYPES OF DATA ANALYTICS

About this module

✓

Data analytics answers questions

✓

Quiz

✓

Quiz Results

EXIT COURSE

Your score 100%

✓

PASSING
66%

TAKE AGAIN

MODULO 3

Modo de revisión

Introduction to Data Concepts

100% COMPLETE

quantitative

About this module

✓

What is big data?

✓

The 5 V's

✓

Quiz

✓

MODULE 3: TYPES OF DATA ANALYTICS

About this module

✓

Data analytics answers questions

✓

Quiz

✓

EXIT COURSE

Added-value contribution

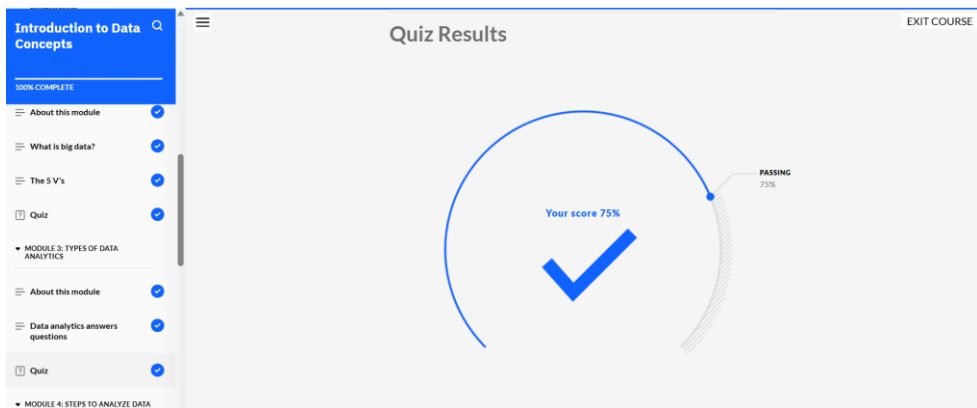
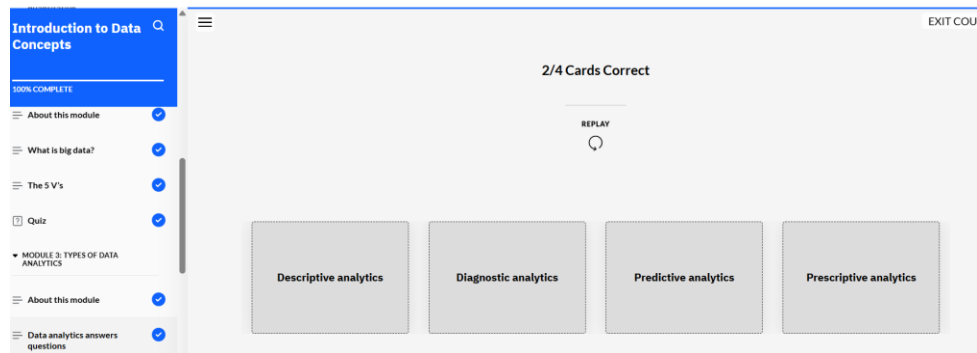
Complexity

Descriptive analytics

Diagnostic analytics

Predictive analytics

Prescriptive analytics



MODULO 4

Introduction to Data Concepts

100% COMPLETE

Quiz

MODULE 4: STEPS TO ANALYZE DATA

About this module

Steps you can easily remember

Walk through with a data analyst

Quiz

MODULE 5: VISUALIZE THE DATA

About this module

It's time to visualize and tell a story

Lesson 16 of 27

EXIT COURSE

Steps you can easily remember

Data analysis is the process of collecting, cleaning, and transforming data to obtain insights to help make better and informed decisions. In our ever-growing, data-driven world, this is a must for companies of all sizes to solve everyday business problems. Each company has its own team, processes, and tools for data analysis projects.

Before beginning a data analysis project, a good starting point is to have a business objective in mind and a clearly defined problem. What questions do you want to answer? After all, 90% of analytics is asking the right questions! You must also determine the metrics to measure performance. What are you measuring and how are you measuring it?

While there is no prescribed data analysis process, these are the typical steps to follow.

MODULO 5

Introduction to Data Concepts

100% COMPLETE

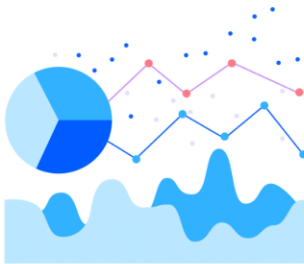
MODULE 5: VISUALIZE THE DATA

- About this module
- It's time to visualize and tell a story
- An early visualization
- Tell a compelling data story
- Types of visualizations
- Select the best chart
- Data can be misleading

EXIT COURSE

What good is data if you can't communicate about it in a way that people understand? This is where visualizations come into play. A data visualization is a graphical display of abstract or complex information. Data analysts use visualizations like charts, graphs, and maps for two reasons:

1. To explore and interpret data during analysis to identify patterns or trends
2. To communicate results and help people understand the insights to make decisions



Introduction to Data Concepts

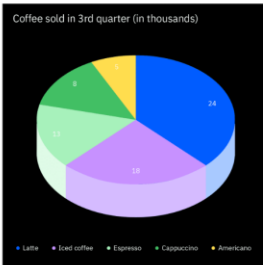
100% COMPLETE

MODULE 5: VISUALIZE THE DATA

- About this module
- It's time to visualize and tell a story
- An early visualization
- Tell a compelling data story
- Types of visualizations
- Select the best chart

EXIT COURSE

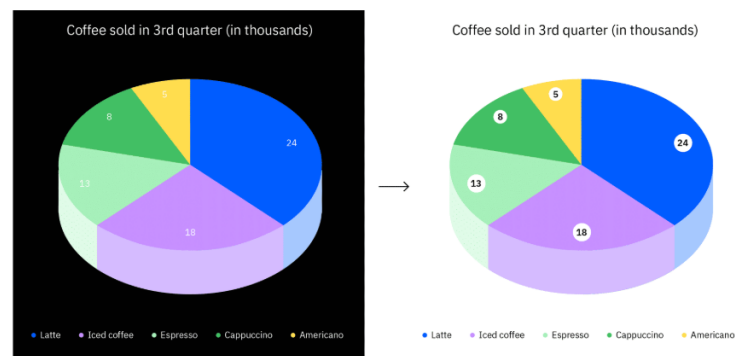
Start with a pie chart



Coffee Type	Sales (in thousands)
Latte	24
Iced coffee	18
Espresso	13
Cappuccino	8
Americano	5

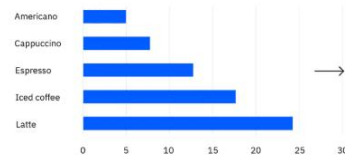
Step 2

Remove the noise

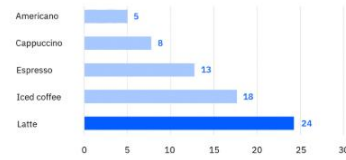


You found the best one

Coffee sold in 3rd quarter (in thousands)



Coffee sold in 3rd quarter (in thousands)



Reflect on this

☒ Reflect

☐ Download your response.

Reflect

El gráfico que hace que Stock se vea mejor dependerá de cómo se presenten los datos visualmente. Algunas razones pueden incluir

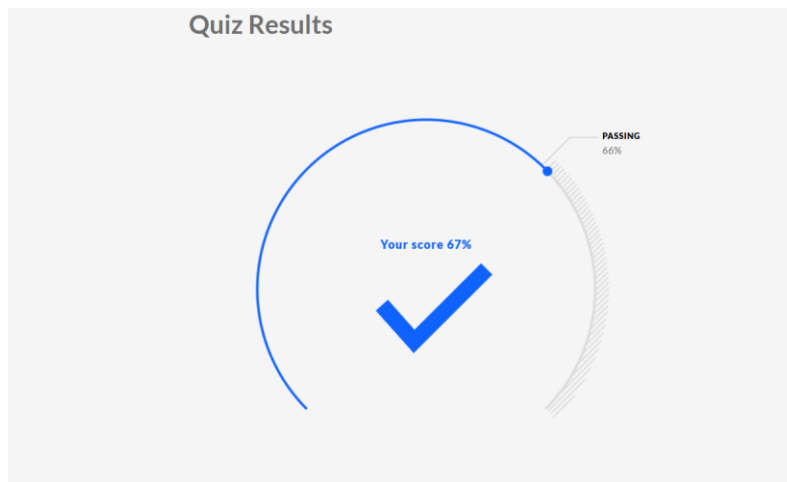
Escala del eje Y: Un gráfico que utilice una escala comprimida en el eje Y puede hacer que los cambios en el valor de Stock parezcan más pequeños y estables.

Tipo de gráfico: Un gráfico de líneas puede resaltar una tendencia positiva a lo largo del tiempo, mientras que un gráfico de barras puede enfocarse en valores específicos que favorecen a la acción.

Remaining characters: 13

<

>



Introduction to Data Concepts

100% COMPLETE

Types of visualizations

Select the best chart

Data can be misleading

Quiz

SUMMARY AND FINAL ASSESSMENT

You've come so far!

Key points to remember

Final assessment

Course completion

Quiz Results

Your score 87%

PASSING
80%

Introduction to Data Concepts

100% COMPLETE

Types of visualizations

Select the best chart

Data can be misleading

Quiz

SUMMARY AND FINAL ASSESSMENT

You've come so far!

Key points to remember

Final assessment

Course completion

Earn a credential!

Reminder: This course is one of several in the Data Fundamentals series. **Complete all of the courses** in the Data Fundamentals learning plan and earn the **Data Fundamentals** digital credential from IBM!

IBM SkillsBuild

EXIT COURSE



For more information about this credential, visit [Data Fundamentals](#) on Credly.

Confirm that course completion shows as **100%** at the top of the navigation menu. If course completion is not 100%, go to any lesson that does not have a checkmark beside it and select the **I've checked it out!** button. You can then close this window to return to SkillsBuild and explore more learning.

Note: It might take a few hours for course completion to appear on SkillsBuild.